# Evaluation runs: **nbuyer** (n-buyer-kragl)

generated 2026-11-08 from brain-data 38584f536 2026-11-08 20:45:11 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | 24 | 300 | brain03, 2026-11-01 |
| 3 | 192 | 192 | conforms | 0.3 | 2 | 1 h, 250 GB | 168 | 15348 | brain03, 2026-11-01 |
| 4 | 1920 | 1920 | conforms | 25.5 | 8 | 1 h, 250 GB | 1344 | 1368282 | brain03, 2026-11-01 |
| 5 | 23040 | 23040 | conforms | 1405.8 | 136 | 1 h, 250 GB | 11819 | 122224448 | brain03, 2026-11-01 |
| 6 | 322560 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-01 |

### stateless: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | 24 | 387 | brain03, 2026-11-01 |
| 3 | 192 | 192 | 96 | violation | 0.6 | 2 | 1 h, 250 GB | 192 | 28380 | brain03, 2026-11-01 |
| 4 | 1920 | 1920 | 1320 | violation | 61.5 | 10 | 1 h, 250 GB | 1920 | 3361944 | brain03, 2026-11-01 |
| 5 | 23040 | 23040 | 18720 | violation | 7083.0 | 132 | 48 h, 250 GB | 23040 | 504590220 | brain04, 2026-11-01 |

**Not run (stateless).** skipped: a smaller size had reached its limit: 6 no-subtract bag, 7 no-subtract bag, 7 correct bag, 8 no-subtract bag, 8 correct bag, 9 correct bag, 9 no-subtract bag, 10 no-subtract bag, 10 correct bag, 11 no-subtract bag, 11 correct bag.

## stateful

### stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | 192 | 192 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 4 | 1920 | 1920 | conforms | 0.2 | 12 | 1 h, 250 GB | brain02, 2026-11-01 |
| 5 | 23040 | 23040 | conforms | 7.9 | 148 | 1 h, 250 GB | brain02, 2026-11-01 |
| 6 | 322560 | 322560 | conforms | 2954.0 | 3227 | 1 h, 250 GB | brain02, 2026-11-01 |
| 7 | | | 🕓 (1 h) | | | 1 h, 250 GB | brain02, 2026-11-01 |

### stateful: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | 192 | 192 | 96 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 4 | 1920 | 1920 | 1320 | violation | 0.3 | 16 | 1 h, 250 GB | brain02, 2026-11-01 |
| 5 | 23040 | 23040 | 18720 | violation | 13.8 | 410 | 1 h, 250 GB | brain02, 2026-11-01 |
| 6 | 322560 | 322560 | 287280 | violation | 11251.0 | 4740 | 48 h, 250 GB | brain03, 2026-11-06 |

**Not run (stateful).** skipped: a smaller size had reached its limit: 7 no-subtract bag, 8 no-subtract bag, 8 correct bag, 9 correct bag, 9 no-subtract bag, 10 no-subtract bag, 10 correct bag, 11 no-subtract bag, 11 correct bag.

left out by the table rules (ok): 6 no-subtract bag.

# offline

## offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 192 | 192 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 1920 | 1920 | conforms | 0.2 | 10 | 10 min, 16 GB | 0.1 | 0.1 | 0.1 | in-memory | brain05, 2026-11-02 |
| 5 | 23040 | 23040 | conforms | 5.7 | 86 | 10 min, 16 GB | 0.6 | 1.0 | 4.6 | in-memory | brain05, 2026-11-02 |
| 6 | 322560 | 322560 | conforms | 1886.7 | 892 | 1 h, 250 GB | 3.9 | 7.9 | 1878.8 | in-memory | brain05, 2026-11-03 |
| 7 | 5160960 | 5160960 | (1 h) | | | 1 h, 250 GB | 27.6 | 60.4 | | in-memory | brain05, 2026-11-03 |
| 8 | 92897280 | 92897280 | (1 h) | | | 1 h, 250 GB | 421.2 | 1070.3 | | in-memory | brain05, 2026-11-03 |

## offline: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-1 |
| 3 | 192 | 192 | 96 | violation | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-1 |
| 4 | 1920 | 1920 | 1320 | violation | 0.3 | 10 | 10 min, 16 GB | 0.1 | 0.2 | 0.1 | in-memory | brain05, 2026-1 |
| 5 | 23040 | 23040 | 18720 | violation | 15.1 | 119 | 10 min, 16 GB | 0.6 | 1.2 | 13.8 | in-memory | brain05, 2026-1 |
| 6 | 322560 | 322560 | 287280 | violation | 12.3 | 277 | 48 h, 250 GB | 3.7 | 8.1 | 4.2 | merge | brain03, 2026-1 |
| 7 | 5160960 | 5160960 | | (1 h) | | | 1 h, 250 GB | 27.6 | 78.7 | | in-memory | brain05, 2026-1 |
| 8 | 92897280 | 92897280 | | oom (250 GB) | | | 1 h, 250 GB | 421.2 | 1459.2 | | in-memory | brain05, 2026-1 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 8 no-subtract bag, 8 correct bag.

skipped: a smaller size had reached its limit: 9 correct bag, 9 no-subtract bag, 10 no-subtract bag, 10 correct bag, 11 no-subtract bag, 11 correct bag.

left out by the table rules (timeout): 6 no-subtract bag.

# cross offline, Java implementation against the Rust specification

## cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.4 | 164 | 10 min, 16 GB | 0.0 | 3.4 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 192 | 192 | conforms | 4.0 | 180 | 10 min, 16 GB | 0.0 | 4.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 1920 | 1920 | conforms | 8.0 | 582 | 10 min, 16 GB | 0.1 | 8.0 | 0.1 | in-memory | brain01, 2026-11-02 |
| 5 | 23040 | 23040 | conforms | 51.5 | 1138 | 10 min, 16 GB | 0.6 | 46.7 | 4.8 | in-memory | brain01, 2026-11-02 |

## cross offline, Java implementation against the Rust specification: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 3.3 | 166 | 10 min, 16 GB | 0.0 | 3.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 192 | 192 | 96 | violation | 4.0 | 159 | 10 min, 16 GB | 0.0 | 4.0 | 0.0 | in-memory | brain01, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1920 | 1920 | 1320 | violation | 7.8 | 605 | 10 min, 16 GB | 0.1 | 7.7 | 0.1 | in-memory | brain01, 2026-11-02 |
| 5 | 23040 | 23040 | 18720 | violation | 58.4 | 1406 | 10 min, 16 GB | 0.6 | 46.5 | 11.9 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Java implementation against the Rust specification).** planned but never scheduled (grid points beyond the ladder): 6 correct bag, 6 no-subtract bag.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.1 | 160 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3 | 192 | 192 | conforms | 3.5 | 184 | 10 min, 16 GB | 3.5 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 4 | 1920 | 1920 | conforms | 5.0 | 244 | 10 min, 16 GB | 5.0 | 0.1 | 0.1 | in-memory | brain01, 2026-11-01 |
| 5 | 23040 | 23040 | conforms | 23.4 | 594 | 10 min, 16 GB | 18.9 | 1.0 | 4.5 | in-memory | brain01, 2026-11-01 |

### cross offline, Rust implementation against the Java specification: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 3.1 | 160 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 3 | 192 | 192 | 96 | violation | 3.5 | 184 | 10 min, 16 GB | 3.5 | 0.1 | 0.0 | in-memory | brain01, 2026-11-01 |
| 4 | 1920 | 1920 | 1320 | violation | 5.1 | 244 | 10 min, 16 GB | 5.0 | 0.2 | 0.1 | in-memory | brain01, 2026-11-01 |
| 5 | 23040 | 23040 | 18720 | violation | 30.9 | 604 | 10 min, 16 GB | 18.9 | 1.2 | 12.1 | in-memory | brain01, 2026-11-01 |

**Not run (cross offline, Rust implementation against the Java specification).** skipped: a smaller size had reached its limit: 6 correct bag, 6 no-subtract bag.

## cross online, Java implementation (JMC) against the Rust specification export

### cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 0.8 | 150 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3 | 192 | 192 | conforms | 1.3 | 172 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 4 | 1920 | 1920 | conforms | 6.7 | 673 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 5 | 23040 | 23040 | conforms | 71.8 | 1562 | 1 h, 250 GB | | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 0.7 | 145 | 1 h, 250 GB | 53 | brain05, 2026-11-02 |
| 3 | 192 | 192 | 96 | violation | 1.2 | 170 | 1 h, 250 GB | 56 | brain05, 2026-11-02 |
| 4 | 1920 | 1920 | 1320 | violation | 6.2 | 681 | 1 h, 250 GB | 59 | brain05, 2026-11-02 |
| 5 | 23040 | 23040 | 18720 | violation | 77.2 | 1229 | 1 h, 250 GB | 64 | brain05, 2026-11-02 |

**Not run (cross online, Java implementation (JMC) against the Rust specification export).** planned but never scheduled (grid points beyond the ladder): 2 correct bag, 2 no-subtract bag, 3 no-subtract bag, 3 correct bag, 4 no-subtract bag, 4 correct bag, 5 no-subtract bag, 5 correct bag.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.1 | 156 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 3 | 192 | 192 | conforms | 3.6 | 180 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 4 | 1920 | 1920 | conforms | 5.2 | 240 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 5 | 23040 | 23040 | conforms | 24.0 | 586 | 1 h, 250 GB | | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 3.1 | 156 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 3 | 192 | 192 | 96 | violation | 3.6 | 180 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 4 | 1920 | 1920 | 1320 | violation | 5.3 | 240 | 1 h, 250 GB | 25 | brain04, 2026-11-01 |
| 5 | 23040 | 23040 | 18720 | violation | 26.7 | 586 | 1 h, 250 GB | 38 | brain04, 2026-11-01 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** left out by the table rules (ok): 2 correct bag, 2 no-subtract bag, 3 no-subtract bag, 3 correct bag, 4 no-subtract bag, 4 correct bag, 5 no-subtract bag, 5 correct bag.

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 322560 | 322560 | conforms | 561.9 | 16265 | 48 h, 250 GB | 559.5 | 7.4 | 2.3 | merge | brain01, 2026-11-04 |

### twins offline, Rust implementation against its Java twin: pair **no-subtract**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 322560 | 322560 | conforms | 569.4 | 17790 | 48 h, 250 GB | 562.8 | 8.0 | 6.6 | merge | brain01, 2026-11-04 |

### twins offline, Rust implementation against its Java twin: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 322560 | 322560 | 287280 | violation | 563.9 | 16419 | 48 h, 250 GB | 559.5 | 8.0 | 4.4 | merge | brain01, 2026-11-04 |

**Not run (twins offline, Rust implementation against its Java twin).** planned but never scheduled (grid points beyond the ladder): 7 bag correct, 7 bag no-subtract.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 322560 | 322560 | conforms | 561.8 | 16265 | 48 h, 250 GB | 7.4 | 559.5 | 2.2 | merge | brain01, 2026-11-04 |

### twins offline, Java implementation against its Rust twin: pair **no-subtract**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 322560 | 322560 | conforms | 569.2 | 17790 | 48 h, 250 GB | 8.0 | 562.8 | 6.4 | merge | brain01, 2026-11-04 |

### twins offline, Java implementation against its Rust twin: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 322560 | 322560 | 287280 | violation | 567.2 | 17636 | 48 h, 250 GB | 7.4 | 562.8 | 4.4 | merge | brain01, 2026-11-04 |

**Not run (twins offline, Java implementation against its Rust twin).** planned but never scheduled (grid points beyond the ladder): 7 bag correct, 7 bag no-subtract.

## twins online, Rust implementation (TraceForge) against the Java twin's export

### twins online, Rust implementation (TraceForge) against the Java twin's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 6 | 322560 | 322560 | conforms | 1866.2 | 16190 | 48 h, 250 GB | | brain01, 2026-11-04 |

### twins online, Rust implementation (TraceForge) against the Java twin's export: pair **no-subtract**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 6 | 322560 | 322560 | conforms | 10421.8 | 17561 | 48 h, 250 GB | | brain01, 2026-11-04 |

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.4 | 162 | 10 min, 16 GB | 3.3 | 0.1 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3 | 192 | 192 | conforms | 4.0 | 158 | 10 min, 16 GB | 4.0 | 0.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | conforms | 7.9 | 573 | 10 min, 16 GB | 7.7 | 0.3 | 0.2 | in-memory | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | conforms | 47.2 | 7292 | 10 min, 16 GB | 42.4 | 1.4 | 4.8 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **no-subtract**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.4 | 158 | 10 min, 16 GB | 3.3 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 192 | 192 | conforms | 4.2 | 167 | 10 min, 16 GB | 4.2 | 0.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | conforms | 7.9 | 569 | 10 min, 16 GB | 7.7 | 0.4 | 0.2 | in-memory | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | conforms | 76.1 | 8321 | 10 min, 16 GB | 42.4 | 1.2 | 33.7 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 3.3 | 162 | 10 min, 16 GB | 3.3 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 192 | 192 | 96 | violation | 4.0 | 158 | 10 min, 16 GB | 4.0 | 0.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | 1320 | violation | 7.8 | 573 | 10 min, 16 GB | 7.7 | 0.4 | 0.1 | in-memory | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | 18720 | violation | 55.4 | 7300 | 10 min, 16 GB | 42.4 | 1.2 | 13.0 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** skipped: a smaller size had reached its limit: 6 correct bag, 6 no-subtract bag.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.3 | 162 | 10 min, 16 GB | 0.1 | 3.3 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 192 | 192 | conforms | 4.0 | 158 | 10 min, 16 GB | 0.1 | 4.0 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | conforms | 7.8 | 573 | 10 min, 16 GB | 0.3 | 7.7 | 0.1 | in-memory | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | conforms | 47.5 | 7292 | 10 min, 16 GB | 1.4 | 42.4 | 5.1 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **no-subtract**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.3 | 158 | 10 min, 16 GB | 0.2 | 3.3 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 192 | 192 | conforms | 4.2 | 167 | 10 min, 16 GB | 0.1 | 4.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | conforms | 7.9 | 569 | 10 min, 16 GB | 0.4 | 7.7 | 0.2 | in-memory | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | conforms | 74.1 | 8321 | 10 min, 16 GB | 1.2 | 42.4 | 31.7 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 3.3 | 158 | 10 min, 16 GB | 0.1 | 3.3 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 192 | 192 | 96 | violation | 4.2 | 167 | 10 min, 16 GB | 0.1 | 4.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | 1320 | violation | 7.8 | 569 | 10 min, 16 GB | 0.3 | 7.7 | 0.1 | in-memory | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | 18720 | violation | 55.5 | 8313 | 10 min, 16 GB | 1.4 | 42.4 | 13.1 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** skipped: a smaller size had reached its limit: 6 correct bag, 6 no-subtract bag.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.9 | 162 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 192 | 192 | conforms | 4.1 | 171 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | conforms | 9.6 | 597 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | conforms | 50.7 | 9196 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-subtract**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.7 | 159 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 192 | 192 | conforms | 4.4 | 170 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | conforms | 10.4 | 645 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | conforms | 49.2 | 10395 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 3.7 | 158 | 1 h, 250 GB | 80 | brain02, 2026-11-03 |
| 3 | 192 | 192 | 96 | violation | 4.9 | 168 | 1 h, 250 GB | 103 | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | 1320 | violation | 10.6 | 2209 | 1 h, 250 GB | 182 | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | 18720 | violation | 47.5 | 9809 | 1 h, 250 GB | 257 | brain02, 2026-11-03 |

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 4.7 | 158 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 192 | 192 | conforms | 4.0 | 154 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | conforms | 7.9 | 569 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | conforms | 48.1 | 7284 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-subtract**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | conforms | 3.4 | 154 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 192 | 192 | conforms | 4.2 | 163 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | conforms | 8.0 | 565 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | conforms | 69.3 | 8305 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-subtract**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 24 | 24 | 6 | violation | 3.3 | 158 | 1 h, 250 GB | 2 | brain02, 2026-11-03 |
| 3 | 192 | 192 | 96 | violation | 4.0 | 154 | 1 h, 250 GB | 2 | brain02, 2026-11-03 |
| 4 | 1920 | 1920 | 1320 | violation | 8.0 | 569 | 1 h, 250 GB | 4 | brain02, 2026-11-03 |
| 5 | 23040 | 23040 | 18720 | violation | 52.7 | 7284 | 1 h, 250 GB | 41 | brain02, 2026-11-03 |